

TA: Ali Hassan

Traffic management system

Team number: twenty-two

ID	Name
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2023170383	عمر أشرف عبدربه مبارك
2023170386	عمر بدر عمر بدر
2023170388	عمر جمال علي حسن
2023170264	سلمى مجدى خيرى حسن
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The Traffic Management System helps manage road traffic by keeping track of vehicles, controlling traffic lights, and recording rule violations. Its main purpose is to reduce traffic jams, make roads safer, and keep traffic flowing smoothly. The system includes features like managing traffic signals, tracking vehicles, recording violations, traffic officers, and vehicle owners.

Input and Output Scenarios

When the system starts, a list of vehicles, owners, violations, officers, zones, and traffic lights is loaded from the files.

Admin Functionalities:

- Add Traffic Lights

Input: List of Zones and list of traffic lights.

Process: Admin inputs data to add and set their durations.

Output: Traffic light is added to the system successfully.

```
Admin@gmail.com
=====
Add a traffic light -> 1
Update a traffic light -> 2
Delete a traffic light -> 3
View violations -> 4
Traffic report -> 5
Add a new officer -> 6
Log out -> 7
1
Enter the zone id you want to add traffic light to :
1
Enter traffic light status (red/yellow/green) :
red
Do you want a default cycle for the traffic light (y/n)?
y
Successful operation.

Welcome Admin.
Admin@gmail.com
```

- Update Traffic Lights

Input: List of Zones and list of traffic lights

Process: Admin inputs data to update.

Output: traffic light new properties.

```
=====
Add a traffic light -> 1
Update a traffic light -> 2
Delete a traffic light -> 3
View violations -> 4
Traffic report -> 5
Add a new officer -> 6
Log out -> 7
2
Enter the zone id you want to update traffic light at :
1
Enter traffic light id :
1
What do you want to update?
Location -> 1
Status -> 2
Duration -> 3
2
Enter the new status :
green
This traffic light id is 1
This traffic light location is borg el-arab
This traffic light status is green
This traffic light status duration is 60 seconds
This traffic light is in a default duration
Successful operation.
```

- Delete Traffic Lights

Process: Admin inputs zone ID and traffic light ID.

Output: Traffic light deleted successfully.

```
=====
Add a traffic light -> 1
Update a traffic light -> 2
Delete a traffic light -> 3
View violations -> 4
Traffic report -> 5
Add a new officer -> 6
Log out -> 7
3
Enter the zone id you want to delete traffic light from :
1
Enter traffic light id :
1
Successful operation.
```

- View Violations by Vehicle or Zone

Input: List of Zones and List of Traffic Violations

Process: Admin requests a list of violations based on vehicle ID or zone ID.

Output: A list of violations is displayed.

```
View violation by vehicle -> 1
View violation by zone id -> 2
View violation by zone name -> 3
2
Enter zone id :
1
Vehcile ID      Violation Type      Fine      Date
2               Driving under the influence  100.0     2024-12-20
=====
Vehcile ID      Violation Type      Fine      Date
1               Speeding              500.0     2024-12-20
=====
Successful operation.
=====
```

- Generate Traffic Reports

Input: list of zones and array list of Traffic Violations

Process: Admin generates reports on high-density zones and frequent violations.

Output: Traffic reports are displayed and can be exported.

```
5
Traffic Report
2024/12/20 05:06:52
borg el-arab,Alex
The frequent violations :
Speeding has been happened 1 times.
Driving under the influence has been happened 1 times.
Reckless driving has been happened 0 times.
Distracted driving has been happened 0 times.
Running a red light has been happened 0 times.
This zone is high density.
=====
new borg el-arab,Alex
The frequent violations :
Speeding has been happened 0 times.
Driving under the influence has been happened 0 times.
Reckless driving has been happened 0 times.
Distracted driving has been happened 0 times.
Running a red light has been happened 0 times.
This zone is high density.
=====
alex city,Alex
The frequent violations :
Speeding has been happened 0 times.
Driving under the influence has been happened 0 times.
Reckless driving has been happened 0 times.
Distracted driving has been happened 0 times.
Running a red light has been happened 0 times.
This zone is high density.
=====
naer city,Cairo
The frequent violations :
Speeding has been happened 0 times.
Driving under the influence has been happened 0 times.
Reckless driving has been happened 0 times.
Distracted driving has been happened 0 times.
Running a red light has been happened 0 times.
This zone is high density.
=====
meadi,Cairo
The frequent violations :
Speeding has been happened 0 times.
Driving under the influence has been happened 0 times.
Reckless driving has been happened 0 times.
Distracted driving has been happened 0 times.
Running a red light has been happened 0 times.
This zone is high density.
=====
```

Traffic Officer Functionalities:

- Calculate fine amounts

Input: violation type

Process: return a fine amount based on the violation type

- Record Violations

Input: violation ID, vehicle ID, violation type, Array List of Vehicles, Array List of all owners

List of Traffic Violations.

Process: The officer enters details of the violation, and the fine is calculated automatically based on the violation type. If the vehicle ID doesn't exist in the system the process stops. A notification method is called to notify the owner. It also sets the vehicle status as not active if it finds more than 2 violations with the same vehicle ID.

Output: The violation is recorded in the system, and a notification is sent to the officer of what he has already recorded and to the owner successfully.

```
*****MENU*****
press -1- View violations recorded by you
press -2- Record violation
press -3- set zone denisty
press -4- Logout
Enter Your Choice: 2
Enter details asked below:
Vehicle ID: 1
Violation Type:
****choices****
1-Speeding
2-Driving under the influence
3-Reckless driving
4-Distracted driving
5-Running a red light
1
=====
*** notification ***
"Speeding violation that id is "3" recorded on 2024-12-20 to vehicle 1. Fine: $500.000"
=====
Notification sent to owner successfully
.....
```

- View Recorded Violations

Input: List of Traffic Violations

Process: The violations are filtered based on the officer ID from the violations file

Output: A list of recorded violations is displayed.

```
*****MENU*****
press -1- View violations recorded by you
press -2- Record violation
press -3- set zone denisty
press -4- Logout
Enter Your Choice: 1
Traffic Violation recorded by officer ali:
Vehcile ID      Violation Type      Fine      Date
1               Speeding              500.0     2024-12-20
-----
Vehcile ID      Violation Type      Fine      Date
2               Driving under the influence  100.0     2024-12-20
-----
Vehcile ID      Violation Type      Fine      Date
1               Speeding              500.0     2024-12-20
```

- Set zone density

Input: y\n

Process: return true if y and if n returns false

```
*****MENU*****:
press -1- View violations recorded by you
press -2- Record violation
press -3- set zone denisty
press -4- Logout
Enter Your Choice: 3
if heigh press y else n
y
done!
```

Vehicle Owner Functionalities:

- Owner logs in and views their violations

Output: Vehicle ID and the violation type and the fine amount and date

```
-----
Welcome, Ziad
1- Add Vehicle
2- Delete Vehicle
3- Check Vehicle Status
4- Show Registered Details
5- Pay Fine
6- View Violations
7- Logout
Enter Your Choice: 6
Vehicle ID      Violation Type      Fine      Date
1               Speeding              500.0     2024-12-20
=====
Notifications for Ziad:
No new notifications.
=====
```

- pay fine

Input: yes/no

Output: fine paid successfully\fine payment cancels

```
-----
Enter your name: Ziad
Enter your ID: 123
Login successful!
=====
Notifications for Ziad:
No new notifications.
=====
-----
Welcome, Ziad
1- Add Vehicle
2- Delete Vehicle
3- Check Vehicle Status
4- Show Registered Details
5- Pay Fine
6- View Violations
7- Logout
Enter Your Choice: 5
Fine Details:
Amount: 500.0
Do you want to pay this fine? (yes/no): yes
1,1,Speeding,2024-12-20,500.0,1,1,Alex,borg el-arab

Fine paid successfully!
```

- Check Vehicle Status

Input: vehicle ID

Process: The owner checks the status

Output: Vehicle status and details are displayed

```
Welcome, Ziad
1- Add Vehicle
2- Delete Vehicle
3- Check Vehicle Status
4- Show Registered Details
5- Pay Fine
6- View Violations
7- Logout
Enter Your Choice: 3
Enter Vehicle ID: 1

Vehicle Status: Active
Do you want to perform another action? (yes/no): no
```

- Registered Details

Input: vehicle ID

Output: display vehicle

```
-----
Welcome, Ziad
1- Add Vehicle
2- Delete Vehicle
3- Check Vehicle Status
4- Show Registered Details
5- Pay Fine
6- View Violations
7- Logout
Enter Your Choice: 4
Enter Vehicle ID: 1
- Owner Name: Ziad
- Vehicle Type: Truck
- License Plate: Abc447
```

- Add vehicle

Input: license plate and vehicle type

Output: added successfully


```
Welcome, Ziad
1- Add Vehicle
2- Delete Vehicle
3- Check Vehicle Status
4- Show Registered Details
5- Pay Fine
6- View Violations
7- Logout
Enter Your Choice: 1
Enter Vehicle Id:
22
Enter vehicle type: (Car ,Truck ,Bike )
car
Enter License Plate:
sdf123
Added Successfully
```

- Delete vehicle

Input: vehicle id

Output: deleted successfully

```
Welcome, Ziad
1- Add Vehicle
2- Delete Vehicle
3- Check Vehicle Status
4- Show Registered Details
5- Pay Fine
6- View Violations
7- Logout
Enter Your Choice: 2
Enter Vehicle ID
22
Deleted Successfully
```

- View notification

Process: Traffic Violation are filtered based on vehicle ID that belongs to the owner and then viewed only once at the beginning if exists and then removed from the notification array list of traffic violations

```
=====
Notifications for Ziad:
Vehicle ID      Violation Type                Fine      Date
1               Speeding                      500.0     2024-12-20
=====
```